

Recovered numerical linear algebra proof drafts

Seven notes develop eight candidate results. Six groups of finite exact-rational checks pass the accompanying verifier. The two row-deletion results are analytic drafts, not computationally certified proofs. Independent mathematical review, source matching, and novelty checks remain necessary.

The IE identifiers below are provisional local labels from recovered notes. The current repository statements and identifiers were not independently retrieved during reconstruction. Each note defines its precise hypotheses; these take precedence over an unverified source mapping.

Note	Topic
IE-13	Unequal-bandwidth partial-pivoting growth
IE-14	Cyclic-tridiagonal partial-pivoting growth
IE-17	LSMR spectral backward-error nonmonotonicity
IE-18	A finite four-step Anderson-acceleration counterexample
IE-19	Entrywise inverse-norm counterexample and infimum
IE-21/22	Random and universal row-deletion constants
IE-23	Nonunique induced-norm-minimizing right inverses

The separate rook-pivoting supplement is partial research and is not counted as a ninth resolution. The archive also contains source code, fresh exact-arithmetic output, a source register, a status ledger, and draft submission notes.

This is a reconstruction, not a bit-for-bit restoration of an earlier archive. A recovered LSMR quadratic-form fraction was corrected against the new exact computation, without changing the strict counterexample. The recovery record gives the details and identifies missing materials.

These documents and their software were developed and reconstructed with substantial AI assistance. No external submission has been made. No claim of peer review, accepted resolution, established novelty, or proof-assistant verification is made.

Sharp partial-pivoting growth for unequal bandwidths

Reconstructed candidate proof draft. Independent mathematical review and source/priority verification are required. The IE identifiers are provisional labels recovered from earlier notes, not verified current repository identifiers.

Abstract

For lower bandwidth $p \geq 1$ and upper bandwidth $q \geq 0$, the proposed optimal dimension-independent GEPP growth factor is h_{p+q} , where $h_t = 0$ for $t \leq 0$ and $h_t = 1 + \sum_{r=1}^p h_{t-r}$ for $t \geq 1$. A sorted-front argument gives an upper bound, and an explicit nonsingular real band matrix attains it. The proof covers fixed column order, real and complex matrices, and arbitrary permitted pivot ties.

1 Statement

Let $A \in \mathbb{R}^{n \times n}$ or $\mathbb{C}^{n \times n}$ be nonsingular, with $a_{ij} = 0$ whenever $i > j + p$ or $j > i + q$. Normalize $\max |a_{ij}| = 1$. Growth includes all active Schur-complement entries, not merely U . Define $G(p, q)$ as the supremum over permitted orders and pivot paths.

Theorem 1. $G(0, q) = 1$. For $p \geq 1$,

$$h_t = 0 \quad (t \leq 0), \quad h_t = 1 + \sum_{r=1}^p h_{t-r} \quad (t \geq 1), \quad \boxed{G(p, q) = h_{p+q}}.$$

A real matrix of order $2p + q + 1$ attains the bound.

The recurrence gives

$$\begin{aligned} G(p, 0) &= 2^{p-1}, \\ G(p, q) &= 2^{p+q-1} - (q-1)2^{q-2} \quad (1 \leq q \leq p+1), \\ G(1, q) &= q+1, \quad G(2, q) = F_{q+4} - 1, \end{aligned}$$

where $F_0 = 0, F_1 = 1$. In particular the equal-bandwidth specialization is $2^{2p-1} - (p-1)2^{p-2}$. These identities follow by induction.

2 Sorted-front lemma

Suppose a front has $p+1$ target-column magnitudes $y_1 \geq \dots \geq y_{p+1} \geq 0$. Remove the row with target magnitude y_a and update the others with multipliers of modulus at most one. The sorted survivors are bounded componentwise by

$$(y_1 + y_2, y_1 + y_3, \dots, y_1 + y_{p+1}).$$

Indeed, the triangle inequality bounds each survivor by $y_a + y_i$. For $i < a$, $y_a + y_i \leq y_1 + y_{i+1}$, since $y_i - y_{i+1} \leq y_1 - y_a$. For $i \geq a$, the ordered bound is $y_a + y_{i+1} \leq y_1 + y_{i+1}$. Sorting and appending a fresh value are monotone. Thus always eliminating the largest target-column

magnitude gives a componentwise upper envelope, even though GEPP actually chooses by another column.

Start with p zeros, append a fresh 1, and perform this envelope update. After one update all survivors equal 1. Subsequently the ordered state evolves by

$$(x_1, \dots, x_p) \mapsto (x_1 + x_2, \dots, x_1 + x_p, x_1 + 1).$$

After $t \geq 1$ updates its i th entry is

$$x_i^{(t)} = 1 + \sum_{r=1}^{p-i+1} h_{t-r},$$

so its largest entry is h_t . Direct substitution proves this identity, including $p = 1$.

3 Upper bound

At the start of stage k , original rows with labels $i \geq k + p$ are unchanged by earlier stages: their first $k - 1$ original entries are zero. They have zero earlier multipliers and cannot be selected as nonzero earlier pivots. Thus there are at most p old surviving rows plus fresh original row $k + p$; later rows are untouched. Near the bottom, missing fresh rows can be padded by zeros.

Fix target column j and set $L = p + q$. For $j \geq L + 1$, all target entries in the front are zero until stage $k_0 = j - L$, when original row $j - q$ arrives. Exactly L updates occur from k_0 through $j - 1$. Each fresh target entry has modulus at most one, so the lemma bounds every value by h_L . For $j \leq L$, the initial p old values are bounded by the canonical state after one imaginary update. At most $j - 1$ actual updates precede elimination of column j , giving $h_j \leq h_L$. Recorded pivot-row entries and untouched entries satisfy the same bound. This covers every active entry. For $p = 0$ the matrix is upper triangular and all elimination multipliers vanish.

4 Attaining construction

For $p \geq 1$, put $j = p + q + 1$, $n = j + p$, and let L_0 be unit lower triangular with -1 on its first p subdiagonals. Use the original-row order

$$\sigma = (p + 1, 1, 2, \dots, p, p + 2, \dots, n), \quad (Pa)_i = a_{\sigma_i}, \quad \eta = 2^{-p}.$$

For $k < j$, define an upper-supported vector $u^{(k)}$ by

$$u_i^{(k)} = \begin{cases} 1 & i = 1, \ k \leq p + 1, \\ 2^{i-2} & 2 \leq i \leq k \leq p + 1, \\ 1 & i = k, \ k \geq p + 2, \\ 0 & \text{otherwise,} \end{cases} \quad A_{:,k} = \eta P^T L_0 u^{(k)}.$$

Set

$$A_{i,j} = \begin{cases} 1 & p + 1 \leq i \leq n, \\ 0 & i \leq p, \end{cases} \quad A_{:,k} = e_k \quad (k > j).$$

For $k \leq p + 1$, the first entry of $L_0 u^{(k)}$ is 1, entries $2, \dots, k$ are zero, and later entries are negatives of sums of subsets of $u^{(k)}$. Their moduli are at most $2^{k-1} \leq 2^p$. After row-label assignment, support lies in original rows $k, \dots, k + p$. For $k \geq p + 2$ the same support follows from $L_0 e_k$. Hence these columns have lower bandwidth at most p and upper bandwidth zero,

with entries bounded by one. The target column starts at $p + 1 = j - q$ and ends at $n = j + p$. Later identity columns respect the band. The initial maximum is one.

In the first $j - 1$ stages in row order σ , the factorization gives multipliers in $\{0, -1\}$ and nonzero pivots. After earlier eliminations, active column k is $L_{0,ik}U_{kk}$; the diagonal multiplier is one, so each chosen pivot is maximal in modulus. Thus these are admissible GEPP choices, not an extra preliminary permutation.

For the target column, $c = PA_{:,j}$ has $c_1 = 1$, $c_i = 0$ for $2 \leq i \leq p + 1$, and $c_i = 1$ for $i \geq p + 2$. Forward substitution gives

$$u_i = c_i + \sum_{r=1}^p u_{i-r}, \quad u_i = 0 \ (i \leq 0).$$

Induction gives $u_1 = 1$ and $u_i = h_{i-1}$ for $i \geq 2$. Therefore the active (j, j) entry at stage j equals h_{p+q} . The leading $j \times j$ block has nonzero pivots, and later columns are identity columns; the matrix is block lower triangular with nonsingular diagonal blocks. This proves attainment.

5 Checks and source status

The standard-library exact verifier checks 60 attaining matrices ($1 \leq p \leq 6$, $0 \leq q \leq 9$), their support, pivot admissibility, growth, and nonsingularity, plus recurrence identities. These finite checks are not a formal proof of all orders.

Recovered target description: *Sharp growth for unequal lower and upper bandwidths*, provisionally IE-13 in OpenProblemsInNLA. The current repository statement could not be retrieved during reconstruction. The result is therefore submitted for review under the precise hypotheses above, not asserted to resolve a source statement whose scope has been independently verified. Repository supplied for the task:¹

¹<https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/linear-systems-and-elimination>.

Sharp growth for cyclic tridiagonal partial pivoting

Reconstructed candidate proof draft. Independent mathematical review and source/priority verification are required. The IE identifiers are provisional labels recovered from earlier notes, not verified current repository identifiers.

Abstract

A two-row front argument gives the proposed sharp GEPP growth $F_{n+1} + 1$ for cyclic tridiagonal matrices of order $n \geq 4$. A rational construction with both cyclic corners nonzero attains it. The argument permits real or complex entries, fixed original column order, and all maximal-modulus pivot ties.

1 Statement

Nonzeros are restricted to the ordinary tridiagonal bands and corners $(1, n)$ and $(n, 1)$, both required nonzero. Matrices are nonsingular, and initial maximum modulus is one. Growth includes every active Schur-complement entry.

Theorem 1. *With $F_0 = 0, F_1 = 1$, the largest permitted growth is*

$$c_n = F_{n+1} + 1 \quad (n \geq 4).$$

The initial values are 6, 9, 14, 22 for $n = 4, 5, 6, 7$.

2 Front estimates

At stage $k \leq n - 2$, two old surviving rows have labels in $\{1, \dots, k\} \cup \{n\}$. Fresh original row $k + 1$ joins them. Other later original rows, except row n , are unchanged because their earlier columns are zero. Eliminating one front row leaves two old rows.

For a target column, let old magnitudes be $a \geq b \geq 0$ and fresh magnitude be at most c . Multipliers have modulus at most one. If $c = 0$, the sorted new pair is bounded by $(a + b, a)$: a zero-valued pivot leaves the old pair unchanged, and either old pivot gives this triangle-inequality bound. For arbitrary c , the survivor sum is at most

$$a + b + c + \max(a, b, c).$$

These statements are also valid over the complex field.

3 All column histories

In the last column, the initial old pair is bounded by $(1, 1)$, and fresh values are zero at stages $1, \dots, n - 3$. After t such updates the pair is bounded by (F_{t+2}, F_{t+1}) . Before stage $n - 2$ it is at most (F_{n-1}, F_{n-2}) . The next fresh value is at most one, so the survivor sum is at most

$$2F_{n-1} + F_{n-2} + 1 = F_{n+1} + 1.$$

The final update uses only these two survivors, so the final scalar and all earlier values satisfy this bound.

Column 1 has only initial values. In column 2, the initial pair is at most $(1, 0)$ and the first fresh value is at most one, giving bound two. For $3 \leq j \leq n - 2$, the old pair in column j is zero until stage $j - 2$. Fresh row $j - 1$ contributes at most one, leaving a pair bounded by $(1, 1)$. At stage $j - 1$ another fresh value at most one gives survivors bounded by two. At the stage eliminating column j , its newly arriving value is at most one. Thus this entire column history is bounded by two.

Column $n - 1$ starts with pair at most $(1, 0)$, because of original row n . Through stage $n - 4$ its fresh values vanish. Before stage $n - 3$ its pair is therefore at most

$$(a, b) = (F_{n-3}, F_{n-4});$$

this includes $n = 4$. A fresh value at most one gives new maximum at most $2a$ and sum at most $2a + b + 1 = F_{n-1} + 1$. At the next stage another fresh value at most one arrives. Each new survivor is bounded by the larger of the old pair sum and old maximum plus one, both at most $F_{n-1} + 1$. This is below the claimed global bound. All columns have now been covered.

4 Attaining construction

Let L be unit lower triangular with -1 on the first two subdiagonals. Define upper triangular U with

$$U_{11} = 1, \quad U_{12} = U_{22} = \frac{1}{2}, \quad U_{ii} = 1 \quad (3 \leq i \leq n - 1),$$

all other nonfinal off-diagonal entries zero, and

$$U_{i,n} = F_{i+1} \quad (i < n), \quad U_{n,n} = F_{n+1} + 1.$$

Put $C = LU$ and assign original rows by

$$C_{i,:} = A_{\sigma_i,:}, \quad \sigma = (1, n, 2, 3, \dots, n - 1).$$

Multiplication gives a cyclic tridiagonal A , with all entry moduli at most one and $A_{1n} = 1$, $A_{n1} = -1$. In particular, the last column of C is $(1, 1, 0, \dots, 0, 1)^T$, by the Fibonacci recurrence. After row assignment, these occupy exactly the allowed rows $1, n - 1, n$. The first two columns follow from the displayed half entries; each remaining nonfinal column comes from a diagonal entry of U and the two subdiagonals of L . This verifies all structural zeros.

The diagonal of U is nonzero. In original-row pivot sequence σ , active column k after earlier elimination consists of $L_{ik}U_{kk}$. Since $L_{kk} = 1$ and $|L_{ik}| \leq 1$, every chosen pivot is maximal in modulus. This describes ordinary permitted GEPP choices, not a preliminary column or row reordering. The final pivot is $F_{n+1} + 1$ and the initial maximum is one, proving attainment.

5 Checks and source status

The verifier constructs and checks the factorization, every structural zero, corner condition, pivot, and intermediate growth for $4 \leq n \leq 30$. The general proof still requires independent mathematical review.

Recovered target description: *Sharp growth for cyclic tridiagonal matrices*, provisionally IE-14 in OpenProblemsInNLA. The live statement and identifier have not been independently verified during reconstruction. Repository supplied for the task:¹

¹<https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/linear-systems-and-elimination>.

An exact spectral backward-error counterexample for LSMR

Reconstructed candidate proof draft. Independent mathematical review and source/priority verification are required. The IE identifiers are provisional labels recovered from earlier notes, not verified current repository identifiers.

Abstract

An integer 4×3 least-squares problem has exact LSMR iterates with increasing optimal matrix-only spectral-norm backward error. A spectral completion gives an upper certificate 1979/2000 for the squared first error, and a positive-definite rational matrix gives a strict lower certificate 99/100 for the squared second error. The stated backward-error approximation increases as well. These certificates use exact rational arithmetic, not a floating-point optimizer or substitution of the Frobenius norm.

1 Definitions and exact iterates

For $r = b - Ax$, define

$$\mu(x) = \min\{\|E\|_2 : (A + E)^T((A + E)x - b) = 0\}.$$

Only A is perturbed; the norm is spectral. The approximation uses $\eta = \|r\|_2 / \|x\|_2$, $K = (A^T, \eta I)^T$, $v = (r^T, 0)^T$, and $\tilde{\mu}(x) = \|KK^\dagger v\|_2 / \|x\|_2$. For nonzero x, r , projection algebra gives

$$\tilde{\mu}(x)^2 = r^T A (\|x\|_2^2 A^T A + \|r\|_2^2 I)^{-1} A^T r.$$

Use no damping, $x_0 = 0$, and

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 5 \\ 0 & 0 & 0 \end{pmatrix}, \quad b = (11, 1, 1, 1)^T.$$

The matrix has full column rank. Put $H = A^T A = \text{diag}(1, 36, 25)$ and $g = A^T b = (11, 6, 5)^T$. Exact LSMR minimizes $\|g - Hx\|_2$ over $\mathcal{K}_k(H, g)$. Its first two iterates are

$$x_1 = \frac{1021}{31201} g = \frac{1}{31201} (11231, 6126, 5105)^T,$$

$$x_2 = \frac{16321g - 383Hg}{110438} = \frac{1}{55219} (87659, 7599, 16865)^T.$$

For $V_1 = (g)$ and $V_2 = (g, Hg)$, direct multiplication checks $x_k \in \text{range } V_k$ and $(HV_k)^T(g - Hx_k) = 0$. The columns of HV_k are independent, so these conditions uniquely certify the minimizing iterates. The third is $(11, 1/6, 1/5)^T$, the exact least-squares solution.

The first residuals are

$$r_1 = \frac{1}{31201} (331980, -5555, 5676, 31201)^T, \quad r_2 = \frac{1}{55219} (519750, 9625, -29106, 55219)^T.$$

The normal residuals decrease, as required by LSMR:

$$\|A^T r_1\|_2^2 = \frac{3593700}{31201} > \frac{5336100}{55219} = \|A^T r_2\|_2^2.$$

2 General lower and upper certificates

For nonzero x , write $s = \|x\|_2^2$, $z = Ax$, $r = b - z$, and

$$C = AA^T, \quad D = \frac{\|r\|_2^2 I - rr^T + zz^T}{s}.$$

Lemma 1. *If $m > n$ and $0 \leq t \leq 1$, then $\mu(x)^2 \geq \lambda_{\min}(tC + (1-t)D)$.*

Proof. For a feasible E whose new residual is nonzero, let u be its unit direction. Then $(A + E)^T u = 0$, so $\|E\|_2^2 \geq u^T C u$. The residual is $uu^T b$, because it is orthogonal to $(A + E)x$. Therefore $Ex = r - uu^T b$, and $\|E\|_2^2 \geq \|r - uu^T b\|_2^2 / s = u^T D u$. A convex combination proves the claim.

For zero new residual, $Ex = r$, so $\|E\|_2^2 \geq \|r\|_2^2 / s$. Choose unit $u \in \ker A^T$, possible since $m > n$. Then $u^T z = 0$ and

$$u^T [tC + (1-t)D] u = (1-t)(\|r\|_2^2 - (u^T r)^2) / s \leq \|r\|_2^2 / s.$$

This proves the same bound in that case. $\square$

Lemma 2. *If a unit vector u has $u^T C u < \kappa$ and $u^T D u < \kappa$, with $\kappa > 0$, then a feasible perturbation has $\|E\|_2 \leq \sqrt{\kappa}$.*

Proof. Let $e = x / \sqrt{s}$ and prescribe

$$E^T u = -A^T u =: de + a_\perp, \quad Ee = (r - uu^T b) / \sqrt{s} =: du + c_\perp,$$

where $a_\perp \perp e$, $c_\perp \perp u$, and the shared scalar is $d = -u^T Ax / \sqrt{s}$. The two squared lengths are $u^T C u, u^T D u$. A completion is

$$E = due^T + ua_\perp^T + c_\perp e^T - \frac{d}{\kappa - d^2} c_\perp a_\perp^T.$$

To bound its norm, put $d_0 = d / \sqrt{\kappa}$, $r_0 = \sqrt{1 - d_0^2}$. In decompositions along u, e , $E / \sqrt{\kappa}$ factors as

$$\begin{pmatrix} 1 & 0 \\ 0 & c_\perp / (\sqrt{\kappa} r_0) \end{pmatrix} \begin{pmatrix} d_0 & r_0 \\ r_0 & -d_0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & a_\perp^T / (\sqrt{\kappa} r_0) \end{pmatrix}.$$

The outside maps have norm at most one by the prescribed length bounds, and the middle map is orthogonal. The prescriptions give $(A + E)^T u = 0$ and $(A + E)x = (I - uu^T)b$, proving feasibility. $\square$

3 Strict increase

For x_1 , choose $w = (250, -1, 1, 27)^T$, $u = w / \sqrt{63231}$, $\kappa = 1979/2000$. Exact evaluation gives

$$u^T C u = \frac{62561}{63231} < \kappa, \quad u^T D u = \frac{1642993919237}{1713779258646} < \kappa.$$

The upper lemma applies. For x_2 , set $t = 5/6$ and $c = 99/100$. Then $tC + (1-t)D - cI = K/2407881992100$, where

$$K = \begin{pmatrix} 206417059721 & -50293465200 & 1125984433750 & -1435003762500 \\ -50293465200 & 83658415217471 & 206242965000 & -26574143750 \\ 1125984433750 & 206242965000 & 61800032332121 & 80360210700 \\ -1435003762500 & -26574143750 & 80360210700 & 11170189945871 \end{pmatrix}.$$

The four leading principal determinants are

$$\begin{aligned} &206417059721, \\ &17265994657467102998545591, \\ &960941324740480331793743845178086291011, \\ &65442104145157248520714038046591467785805073713081. \end{aligned}$$

All are positive. Sylvester's criterion and the lower lemma give

$$\mu(x_1)^2 \leq \frac{1979}{2000} < \frac{99}{100} < \mu(x_2)^2.$$

4 Approximation and rational witness

The squared approximations are exactly

$$\begin{aligned} \tilde{\mu}(x_1)^2 &= \frac{69694107852573439503892031925}{69323394392991282508138323472}, \\ \tilde{\mu}(x_2)^2 &= \frac{5430772101137459612205263871781350}{5387955615790281743396033884265233}. \end{aligned}$$

Cross multiplication yields

$$\tilde{\mu}(x_1)^2 < 1.006 < 1.007 < \tilde{\mu}(x_2)^2,$$

where the decimals are exact rational cutoffs.

The upper completion also has a fully rational form. Put $\omega = w^T w$, $h = w^T z$, $c_0 = r - w(w^T r)/\omega$, $a_0 = -A^T w + (h/s)x$. Then

$$E = -ww^T A/\omega + c_0 x^T/s + \frac{h c_0 a_0^T}{\omega s \kappa - h^2}.$$

The verifier directly checks the perturbed normal equations and positivity of all leading principal minors of $\kappa I - E^T E$. The complete rational matrix is saved in its results file.

A dense variant multiplies A, b by the Hadamard matrix with rows $(1, 1, 1, 1)$, $(1, -1, 1, -1)$, $(1, 1, -1, -1)$, $(1, -1, -1, 1)$, producing

$$A' = \begin{pmatrix} 1 & 6 & 5 \\ 1 & -6 & 5 \\ 1 & 6 & -5 \\ 1 & -6 & -5 \end{pmatrix}, \quad b' = (14, 10, 10, 10)^T.$$

This multiplier is twice orthogonal. Iterates are unchanged and both error quantities double, preserving the strict increases without a zero row.

5 Sources and scope

Recovered references: OpenProblemsInNLA, provisionally IE-17, *Monotonic optimal backward error along LSMR*; D. C.-L. Fong and M. A. Saunders, *LSMR: An Iterative Algorithm for Sparse Least-Squares Problems*, SIAM J. Sci. Comput. 33(5), 2011, 2950–2971, DOI 10.1137/10079687X.¹ The target addressed here uses the explicit spectral-norm definition in Section 1. The current repository statement was not independently retrieved during reconstruction. A different norm or a perturbation model allowing changes to b is outside this certificate's stated claim.

¹<https://arxiv.org/abs/1006.0758>. Reference metadata was retained from earlier notes; source conventions and priority require verification.

Failure of a pairwise four-step bound for restarted Anderson acceleration

Reconstructed candidate proof draft. Independent mathematical review and source/priority verification are required. The IE identifiers are provisional labels recovered from earlier notes, not verified current repository identifiers.

Abstract

Exact diagonal examples refute a proposed pairwise formula for four-step residual amplification. One example has both the iteration matrix and its complement positive definite. A positive-definite one-parameter family gives an unbounded ratio of actual amplification to the proposed expression. These finite-step statements do not resolve the separate asymptotic convergence-factor conjecture.

1 The finite-step assertion

For symmetric M with $A = I - M$ nonsingular, define

$$\alpha(v) = \frac{v^T A v}{v^T A^2 v}, \quad R(v) = M[I - \alpha(v)A]v \quad (v \neq 0), \quad R(0) = 0.$$

The residual after four original steps is $R(R(v))$. For eigenvalues m_i , let

$$\Lambda(M) = \max_{i < j} \left(\frac{m_i m_j (m_j - m_i)}{|m_i(1 - m_i)| + |m_j(1 - m_j)|} \right)^2.$$

For the formula to be defined, each pair considered must have nonzero denominator; all pairs in the examples below do. The target assertion as recovered is $\max_{v \neq 0} \|R(R(v))\|_2 / \|v\|_2 = \Lambda(M)$. The square is part of the proposed norm amplification; squared-norm comparisons require $\Lambda(M)^2$.

2 Exact examples

For $M = \text{diag}(1/10, 1/2, 3/5)$ and $v = (1, 1, 1)^T$, direct rational substitution gives

$$\alpha(v) = 90/61, \quad R(v) = (-2, 8, 15)^T/61, \quad \alpha(R(v)) = 3140/1381,$$

$$R(R(v)) = (289, -756, 1125)^T/84241, \quad \Lambda(M) = 1/121.$$

Consequently

$$\frac{\|R(R(v))\|_2^2}{\|v\|_2^2} = \frac{1920682}{21289638243} > \frac{1}{14641} = \Lambda(M)^2,$$

because $1920682 \cdot 14641 = 28120705162 > 21289638243$. Both M and $I - M$ are positive definite, and $\rho(M) = 3/5 < 1$.

For the simpler semidefinite $M = \text{diag}(0, 1/2, 2/3)$,

$$\alpha(v) = 66/49, \quad R(v) = (0, 8, 18)^T/49, \quad \alpha(R(v)) = 35/13,$$

$$R(R(v)) = (0, -18, 16)^T/637, \quad \Lambda(M) = 4/289.$$

Here the squared comparison is $580/1217307 > 16/83521$, certified by $48442180 > 19476912$ after cross multiplication. No optimization or tolerance is involved.

3 Unbounded underestimation

Let $M_\varepsilon = \text{diag}(\varepsilon^2, 1/2, 1/2 + \varepsilon)$, $v = (1, 1, 1)^T$, with $0 < \varepsilon < 1/2$. Both M_ε and its complement are positive definite. We claim

$$\frac{\|R_\varepsilon(R_\varepsilon(v))\|_2}{\|v\|_2} = \frac{\varepsilon}{6\sqrt{6}} + O(\varepsilon^2), \quad \Lambda(M_\varepsilon) = \frac{\varepsilon^2}{4} + O(\varepsilon^3).$$

All intermediate denominators are nonzero at $\varepsilon = 0$. Expansion gives

$$\alpha_\varepsilon(v) = \frac{4}{3} + \frac{2}{9}\varepsilon + O(\varepsilon^2).$$

For $w_\varepsilon = R_\varepsilon(v)$,

$$w_0 = (0, 1/6, 1/6)^T, \quad w'_0 = (0, -1/18, 17/18)^T.$$

With $a(\varepsilon) = (1 - \varepsilon^2, 1/2, 1/2 - \varepsilon)$, the second coefficient is

$$\beta_\varepsilon = \frac{\sum_i a_i w_{\varepsilon,i}^2}{\sum_i a_i^2 w_{\varepsilon,i}^2} = 2 + 2\varepsilon + O(\varepsilon^2).$$

Its numerator has value $1/36$ and derivative $13/108$ at zero, and denominator has value $1/72$ and derivative $5/108$. Differentiating the second residual gives

$$R_\varepsilon(w_\varepsilon) = \varepsilon(0, -1/12, 1/12)^T + O(\varepsilon^2),$$

proving the norm expansion. The pair $1/2, 1/2 + \varepsilon$ contributes

$$\left(\frac{\varepsilon(1 + 2\varepsilon)}{2 - 4\varepsilon^2} \right)^2 = \varepsilon^2/4 + O(\varepsilon^3).$$

Pairs involving ε^2 contribute $O(\varepsilon^4)$, since their denominators tend to $1/4$. Thus the former pair is maximal for sufficiently small ε , proving the second expansion. Their ratio is

$$\boxed{\frac{2}{3\sqrt{6}}\varepsilon^{-1} + O(1) \longrightarrow \infty.}$$

No fixed multiplicative constant repairs the proposed expression as a uniform four-step upper bound on positive-definite contractions.

4 Independent recurrence check and limits

The code independently executes two cycles of

$$r_1 = Mr_0, \quad d = r_1 - r_0, \quad \gamma = -r_1^T d / (d^T d), \quad r_2 = M(r_1 + \gamma d).$$

Since $d = -Ar_0$, this gives $r_2 = R(r_0)$. The exact vectors and inequalities are checked in both formulations. The continuous-family result is analytic, not established by finite sampling. A large amplification at one residual need not persist on an orbit; the asymptotic-convergence conjecture remains outside these results.

5 Sources and scope

Recovered references: OpenProblemsInNLA, provisionally IE-18, *Exact four-step amplification for restarted Anderson acceleration*; O. A. Krzysik, H. De Sterck and A. Smith, *Asymptotic convergence of restarted Anderson acceleration for certain normal linear systems*, SIAM J. Sci. Comput. 47 (2025), DOI 10.1137/24M1672262, arXiv:2312.04776v4, equation (27).¹ The finite-step assertion in Section 1 is the precise mathematical target here. No assertion is made that a different formula, exponent convention, or asymptotic statement in the original paper is refuted without verifying that source.

¹<https://arxiv.org/html/2312.04776v4>. Bibliographic details and exact target scope are retained as source leads, not independently verified during reconstruction.

An entrywise inverse-norm conjecture and its sharp replacement

Reconstructed candidate proof draft. Independent mathematical review and source/priority verification are required. The IE identifiers are provisional labels recovered from earlier notes, not verified current repository identifiers.

Abstract

A 3×3 positive strictly diagonally dominant matrix refutes the displayed entrywise inverse-norm inequality. Under the stated entrywise upper bounds, the sharp inverse infinity-norm infimum is $1/(\alpha+m)$, rather than the value at $S = \alpha I + m\mathbf{1}\mathbf{1}^T$. The conclusion concerns entrywise ordering, not a stronger order also comparing diagonal-dominance margins.

1 Counterexample

Let $n \geq 3$, $m > 0$, $\alpha \geq (n-2)m$, and $S = \alpha I + m\mathbf{1}\mathbf{1}^T$. The target as recovered asks whether a symmetric, entrywise positive, diagonally dominant $J \leq S$ entrywise must have $\|J^{-1}\|_\infty \geq \|S^{-1}\|_\infty$.

For $n = 3$, $m = \alpha = 1$, take

$$S = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \quad J = \begin{pmatrix} 2 & 1/2 & 1/2 \\ 1/2 & 2 & 1/2 \\ 1/2 & 1/2 & 2 \end{pmatrix}.$$

All entries of J are positive, $J \leq S$ entrywise, and J is strictly diagonally dominant. Yet

$$S^{-1} = I - \frac{1}{4}\mathbf{1}\mathbf{1}^T, \quad J^{-1} = \frac{2}{3}I - \frac{1}{9}\mathbf{1}\mathbf{1}^T, \quad \boxed{\|J^{-1}\|_\infty = \frac{7}{9} < \frac{5}{4} = \|S^{-1}\|_\infty}.$$

2 Sharp infimum

Theorem 1. *For every admissible n, m, α , over the stated class,*

$$\boxed{\inf_J \|J^{-1}\|_\infty = \frac{1}{\alpha + m}}.$$

Strict positivity of all entries prevents attainment.

Proof. All such J are positive definite, because

$$x^T J x = \sum_i \left(J_{ii} - \sum_{j \neq i} J_{ij} \right) x_i^2 + \sum_{i < j} J_{ij} (x_i + x_j)^2.$$

Every term is nonnegative, and for $n \geq 3$, vanishing of all $x_i + x_j$ forces $x = 0$. Cauchy–Schwarz applied to $J^{1/2}e_i, J^{-1/2}e_i$ gives $J_{ii}(J^{-1})_{ii} \geq 1$. Equality would require e_i to be an eigenvector, impossible with positive off-diagonal entries. With $D = \alpha + m$,

$$\|J^{-1}\|_\infty \geq (J^{-1})_{ii} > 1/J_{ii} \geq 1/D.$$

For $0 < \varepsilon \leq m$, let $J_\varepsilon = (D - \varepsilon)I + \varepsilon \mathbf{1}\mathbf{1}^T$. Its diagonal is D , its off-diagonal entries are ε , and $D \geq (n - 1)m \geq (n - 1)\varepsilon$, so it belongs to the class. Direct inversion yields

$$\|J_\varepsilon^{-1}\|_\infty = \frac{D + (2n - 3)\varepsilon}{(D - \varepsilon)(D + (n - 1)\varepsilon)} \longrightarrow \frac{1}{D}.$$

This proves both the value and nonattainment. $\square$

The gap between the proposed value and the actual infimum is

$$\|S^{-1}\|_\infty - \frac{1}{\alpha + m} = \frac{(n - 1)m(\alpha + 2m)}{\alpha(\alpha + m)(\alpha + nm)} > 0.$$

3 Scope, checks, and sources

The exact verifier checks both inverses and the class assumptions, and tests the parameterized inverse formula at rational parameters. The universal lower bound and limiting claim are analytic arguments. A stronger ordering involving dominance margins is a different question; this example is not claimed to satisfy it.

Recovered references: OpenProblemsInNLA, provisionally IE-19, *Sharp inverse norm bound from upper bounds on matrix entries*; originating preprint arXiv:1203.6812, Conjecture 8.1.¹ The current repository statement was not retrieved. The result concerns the explicit formulation in Section 1 and must not be attributed to a different formulation without checking the source.

¹<https://arxiv.org/abs/1203.6812>. Bibliographic details and original-source conventions are recovered reference leads, not independently verified here.

Sharp Gaussian constants for row deletion in unit-row matrices

Reconstructed candidate proof draft. Independent mathematical review and source/priority verification are required. The IE identifiers are provisional labels recovered from earlier notes, not verified current repository identifiers.

Abstract

Analytic proof drafts give the Gaussian trimmed-second-moment limit for adversarial row retention in independent spherical matrices whenever $n \rightarrow \infty$ and $m/n \rightarrow \infty$, with explicit probability bounds. A projection argument proposes the same sharp universal asymptotic upper constant over all unit-row matrices, without an aspect-ratio requirement for the upper bound. These are analytic results; the accompanying finite verifier does not certify them.

1 Definitions and statements

Fix $0 < \theta < 1$, let $A \in \mathbb{R}^{m \times n}$ have unit Euclidean row norms, and set $k = \lfloor \theta m \rfloor$. Define

$$s_\theta(A) = \min_{|S|=k} \sigma_{\min}(A_S) = \min_{\|x\|_2=1} \left(\sum_{i=1}^k (|Ax|^2)_{(i)} \right)^{1/2},$$

where order is increasing, $\sigma_{\min}$ denotes the variational minimum over unit vectors (zero for a rank-deficient map), and empty sums are zero. The equality follows by interchanging the finite and compact minima.

Let $G \sim N(0, 1)$, $\mathbb{P}(|G| \leq a_\theta) = \theta$, and let φ be its density. Set

$$h_\theta = \mathbb{E}[G^2 \mathbf{1}_{|G| \leq a_\theta}] = \theta - 2a_\theta \varphi(a_\theta), \quad c_\theta = \sqrt{h_\theta}.$$

Theorem 1 (Random rows). *For independent uniform rows on S^{n-1} , along every sequence $n \rightarrow \infty$, $m/n \rightarrow \infty$, in probability*

$$\frac{n}{m} s_\theta(A)^2 \rightarrow h_\theta, \quad \frac{n}{m} \|A\|_2^2 \rightarrow 1, \quad \frac{s_\theta(A)^2}{\|A\|_2^2} \rightarrow h_\theta.$$

Theorem 2 (Universal constant). *Uniformly in all integers $m \geq 1$ and unit-row matrices $A \in \mathbb{R}^{m \times n}$,*

$$\sqrt{n/m} s_\theta(A) \leq c_\theta + o_{n \rightarrow \infty}(1).$$

No smaller constant has the eventual uniform upper-bound property. Along every $n \rightarrow \infty$, $m/n \rightarrow \infty$, the supremum over unit-row matrices of $\sqrt{n/m} s_\theta(A)$ converges to c_θ .

2 Trimming identities

For nonnegative $y_1, \dots, y_m$,

$$\frac{1}{m} \sum_{i=1}^k y_{(i)} = \max_{t \geq 0} \left\{ \frac{k}{m} t - \frac{1}{m} \sum_i (t - y_i)_+ \right\}.$$

The right side is concave piecewise linear, with slope k/m minus the empirical distribution function, and is maximized between the k th and $(k+1)$ st order statistics; use $t = 0$ for $k = 0$.

For a nonnegative random variable Y , define

$$H_\theta(Y) = \inf\{\mathbb{E}(Yw) : 0 \leq w \leq 1, \mathbb{E}w = \theta\}.$$

This is the integral of the lower quantiles through mass θ . For a continuous distribution it is $\mathbb{E}[Y\mathbf{1}_{Y \leq b_\theta}]$. Under any coupling of Y, Z ,

$$|H_\theta(Y) - H_\theta(Z)| \leq \mathbb{E}|Y - Z|.$$

Indeed use the same feasible selectors on the joint space. Allowing selectors to depend on both variables does not change either infimum: conditional expectation given one variable preserves the constraints.

3 Pointwise concentration

Let u be uniform on S^{n-1} , $n \geq 2$, $Y_n = nu_1^2$, and $h_{\theta,n} = H_\theta(Y_n)$. Its law is continuous and its mean is one. Put $L = 2/(1 - \theta)$. For independent copies Y_i , $T_m = m^{-1} \sum_{i=1}^m Y_{(i)}$, and $0 < \epsilon \leq (1 - \theta)/2$, we claim

$$\mathbb{P}(|T_m - h_{\theta,n}| > 2L\epsilon + L/m) \leq 5e^{-2m\epsilon^2}.$$

Let b be the θ -quantile. Since the mean is one, $b \leq 1/(1 - \theta) \leq L$. Hoeffding's bounded-variable inequality gives, outside total probability $4e^{-2m\epsilon^2}$,

$$\left| m^{-1} \sum_i \mathbf{1}_{Y_i \leq b} - \theta \right| \leq \epsilon, \quad \left| m^{-1} \sum_i Y_i \mathbf{1}_{Y_i \leq b} - h_{\theta,n} \right| \leq b\epsilon.$$

Markov's inequality gives $\mathbb{P}(Y_n \leq L) \geq (1 + \theta)/2$. The chance fewer than k samples are at most L is at most $e^{-m(1-\theta)^2/2} \leq e^{-2m\epsilon^2}$. On the complementary events, replacing the sum below b by the k smallest terms changes at most $m\epsilon + 1$ terms, each at most L . This proves the claim. For $k = 0$, use $h_{\theta,n} \leq \theta < 1/m$.

The bounded-variable inequality used here follows from $\mathbb{E}e^{\lambda(X - \mathbb{E}X)} \leq e^{\lambda^2 b^2/8}$ for $0 \leq X \leq b$, independence, and optimization of the Chernoff parameter. The scalar exponential bound follows by convexity on an interval and a one-variable maximization.

4 Covariance and uniform directions

The moments satisfy

$$\mathbb{E}Y_n^r = \frac{n^r(2r-1)!!}{n(n+2)\cdots(n+2r-2)} \leq 2^r r!,$$

so $\mathbb{E}|Y_n - 1|^r \leq 4^r r!$ and, for $|\lambda| \leq 1/8$, $\mathbb{E}e^{\lambda(Y_n - 1)} \leq e^{32\lambda^2}$. The mean-zero linear term vanishes; summing the remaining geometric series gives this bound. Chernoff's inequality then yields, for $0 < u \leq 1$,

$$\mathbb{P}\left(\left|m^{-1} \sum_i Y_i - 1\right| > u\right) \leq 2e^{-mu^2/128}.$$

A $1/4$ -net of the real unit sphere has at most 9^n points; a symmetric matrix's norm is at most twice its maximum absolute quadratic form on that net. For $C = (n/m)A^T A$ this gives

$$\mathbb{P}(\|C - I\|_2 > t) \leq 29^n e^{-mt^2/512} \quad (0 < t \leq 1).$$

The net cardinality follows by disjoint-ball volume comparison; the quadratic-form estimate follows by approximating a maximizing unit vector.

Let $T(x) = (n/m) \min_{|S|=k} \|A_S x\|_2^2$. On $\|C - I\|_2 \leq t$, every submatrix quadratic form has norm at most $1 + t$, hence

$$|T(x) - T(y)| \leq 2(1 + t) \|x - y\|_2$$

for unit x, y . A δ -net has at most $(1 + 2/\delta)^n$ points. Pointwise concentration, a union bound, and this Lipschitz estimate give

$$\sup_{\|x\|_2=1} |T(x) - h_{\theta,n}| \leq D := 2L\epsilon + L/m + 2(1 + t)\delta$$

except with probability

$$29^n e^{-mt^2/512} + 5(1 + 2/\delta)^n e^{-2m\epsilon^2},$$

where $0 < t < 1$, $0 < \delta < 1$, $0 < \epsilon \leq (1 - \theta)/2$.

5 Gaussian limit and explicit random error

Couple u with an independent radius $R^2 \sim \chi_n^2$. Then Ru is standard Gaussian, so $G_1^2 = (R^2/n)Y_n$. Independence and the chi-square variance give

$$\mathbb{E}|Y_n - G_1^2| = \mathbb{E}|1 - R^2/n| \leq \sqrt{2/n}, \quad |h_{\theta,n} - h_\theta| \leq \sqrt{2/n}.$$

With the preceding exceptional probability,

$$\begin{aligned} \left| \frac{n}{m} s_\theta(A)^2 - h_\theta \right| &\leq D + \sqrt{2/n}, \\ \left| \frac{s_\theta(A)^2}{\|A\|_2^2} - h_\theta \right| &\leq \frac{D + \sqrt{2/n} + t}{1 - t}. \end{aligned}$$

Here $h_\theta \leq 1$. If $Q = m/n \rightarrow \infty$, eventually choose $t = \epsilon = \delta = 32\sqrt{\log Q/Q}$. Parameter restrictions eventually hold; both failure probabilities vanish and the error is

$$O_\theta \left(\sqrt{\frac{n}{m} \log \frac{m}{n}} + n^{-1/2} + m^{-1} \right).$$

No additional condition on how quickly Q diverges is needed. This proves the random theorem.

6 Deterministic projection argument

For arbitrary unit-row A , choose $1 \leq r < n$, set $d = n - r$, and remove the r largest covariance eigenvector directions. In orthonormal coordinates on the remaining subspace, call the projected matrix $B \in \mathbb{R}^{m \times d}$. Since $\text{tr}(A^T A) = m$,

$$\|B\|_2^2 \leq m/(r + 1), \quad \|b_i\|_2 \leq 1, \quad \text{tr}(B^T B) \leq m.$$

Let $g \sim N(0, I_d)$ and $Y_i = (b_i^T g)^2$; these variables may be correlated. Define

$$\Psi_t(g) = \frac{k}{m}t - \frac{1}{m} \sum_i (t - Y_i)_+.$$

Each scalar Gaussian $b_i^T g$ has variance at most one, so

$$\mathbb{E}\Psi_t(g) \leq \theta t - \mathbb{E}(t - G^2)_+ \leq h_\theta.$$

The final inequality is the population trimming formula, maximized at $t = a_\theta^2$.

Gaussian Poincaré gives $\text{Var}(f(g)) \leq \mathbb{E} \|\nabla f(g)\|_2^2$. A proof expands smooth functions in tensor Hermite polynomials: variance sums squared nonconstant coefficients, while the gradient norm weights them by total degree, at least one. Smooth approximation extends the inequality to Lipschitz functions. Since the scalar derivative of $(t - z^2)_+$ is bounded by $2\sqrt{t}$,

$$\text{Var}(\Psi_t(g)) \leq \frac{\|B\|_2^2}{m^2}(4tm) \leq \frac{4t}{r+1}.$$

This handles arbitrary row correlations without asserting independence.

7 Explicit uniform bound and sharpness

For $0 < \delta < 1$, a grid of $[0, L]$ with spacing at most δ has at most $L/\delta + 2$ points. Chebyshev's inequality and the last variance bound show that all grid values obey $\Psi_t \leq h_\theta + \delta$, except with probability at most

$$\frac{4L(L/\delta + 2)}{(r+1)\delta^2}.$$

Since $t \mapsto \Psi_t$ is 1-Lipschitz, its supremum on $[0, L]$ is at most $h_\theta + 2\delta$ on that event's complement.

Also $Z = m^{-1} \|Bg\|_2^2$ has mean at most one and variance $2\text{tr}((B^T B)^2)/m^2 \leq 2/(r+1)$. Thus $Z \leq 2$ except with probability $2/(r+1)$. On that event, the k th order statistic is at most L , because at least $m - k + 1 \geq (1 - \theta)m$ terms are as large. Therefore the full trimming maximum is attained in $[0, L]$; the case $k = 0$ is immediate.

Finally, $\mathbb{P}(\|g\|_2^2 < (1 - \delta)d) \leq 2/(d\delta^2)$. Consequently, if

$$\frac{2}{r+1} + \frac{4L(L/\delta + 2)}{(r+1)\delta^2} + \frac{2}{(n-r)\delta^2} < 1,$$

there exists a realization satisfying all the bounds. Normalize that g to a unit direction in the retained subspace. The trimming identity yields

$$\boxed{\frac{n}{m} s_\theta(A)^2 \leq \frac{n}{(n-r)(1-\delta)} (h_\theta + 2\delta).}$$

All constants are independent of m, A . Taking $r = \lfloor n^{2/3} \rfloor$, $\delta = n^{-1/6}$ makes the condition hold for all sufficiently large n for fixed θ ; its largest term is $O_\theta(n^{-1/6})$. The result is the uniform bound $(n/m) s_\theta(A)^2 \leq h_\theta + O_\theta(n^{-1/6})$.

The random theorem supplies deterministic realizations arbitrarily close to h_θ along every high-aspect-ratio sequence. Thus no smaller constant can satisfy the eventual uniform requirement, and the normalized supremum converges to c_θ . This proves the second theorem's two quantifier claims without interchanging an unjustified random minimum and expectation.

8 Sources and verification limits

Recovered target descriptions: *Sharp row-deletion constant for spherical random matrices*, provisionally IE-21, and *Optimal uniform row-deletion constant*, provisionally IE-22, in OpenProblemsInNLA.¹ Earlier conversational summaries assigned different identifiers to these topics; the present labels are local identifiers only until the repository mapping is checked.

The live source statements could not be retrieved during reconstruction. The precise normalization, retention convention, singular-value definition, and quantifiers in Section 1 are essential. The probability and existence arguments need independent mathematical review; finite tests of the other notes do not certify them.

¹<https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/linear-systems-and-elimination>.

Nonuniqueness of induced-norm-minimizing right inverses

Reconstructed candidate proof draft. Independent mathematical review and source/priority verification are required. The IE identifiers are provisional labels recovered from earlier notes, not verified current repository identifiers.

Abstract

The same 2×3 full-row-rank matrix has multiple right inverses minimizing its induced $\ell^p \rightarrow \ell^2$ norm for every $2 < p < \infty$, over both fields. The complete minimizer set is a disk over the complex field and an interval over the reals. A complex higher-dimensional construction gives a ball of minimizers. The smallest matrix is attributed in the recovered notes to the motivating paper's spectral-norm example; its construction is not claimed as new.

1 Pointwise orthogonality

For full-row-rank $A \in \mathbb{C}^{m \times n}$, $m < n$, consider

$$\min_{AY=I_m} \|Y\|_{p \rightarrow 2}, \quad \|Y\|_{p \rightarrow 2} = \sup_{z \neq 0} \|Yz\|_2 / \|z\|_p.$$

If $B = A^\dagger$, every right inverse is $Y = B + N$ with $AN = 0$. Since $B = A^*(AA^*)^{-1}$, $B^*N = 0$, and therefore

$$\|Yz\|_2^2 = \|Bz\|_2^2 + \|Nz\|_2^2.$$

Thus B minimizes the objective for every p . This is not the different product objective $\|YA\|_{p \rightarrow 2}$.

2 Smallest counterexample

Theorem 1. *Let*

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad B = \frac{1}{3} \begin{pmatrix} 1 & 1 \\ 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Then $B = A^\dagger$, $AB = AX = I_2$, $B \neq X$, and for $2 \leq p \leq \infty$,

$$\|B\|_{p \rightarrow 2} = \|X\|_{p \rightarrow 2} = \min_{AY=I_2} \|Y\|_{p \rightarrow 2} = 2^{1/2-1/p}.$$

Proof. Direct multiplication gives $B^*B = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \preceq I_2$ and $X^*X = I_2$. The inequality $\|z\|_2 \leq 2^{1/2-1/p} \|z\|_p$ gives both upper bounds. At $z = (1, -1)^T$ both matrices preserve Euclidean norm, so the bounds are attained. Pointwise orthogonality proves global optimality. $\square$

When $m = 1$, the right inverse is a single column, and its induced norm is its Euclidean norm for all p ; orthogonality makes the minimum unique. Hence dimensions 2×3 are smallest.

3 Complete minimizer set

Every right inverse is $B + v(a, b)$, where $v = (-1, 1, 1)^T$. At norming vector $(1, -1)^T$, orthogonality forces $a = b = t$ for any minimizer. The Gram matrix of $Y = B + tv(1, 1)$ has eigenvalue one on $(1, -1)^T$ and eigenvalue $1/3 + 6|t|^2$ on $(1, 1)^T$. Both vectors have equal-modulus coordinates. Testing the latter forces $|t| \leq 1/3$. Conversely this condition gives $Y^*Y \preceq I_2$ and the same attained bound. Thus for every $2 \leq p \leq \infty$, all minimizers are

$$\boxed{B + t(-1, 1, 1)^T(1, 1), \quad |t| \leq \frac{1}{3}.$$

The real parameter interval is $[-1/3, 1/3]$, and X corresponds to $t = 1/3$.

4 Every complex dimension

For $2 \leq m < n$, take $A = (\mathbf{1} \ I_m \ 0_{m \times (n-m-1)})$, $B = A^\dagger$. Then

$$B^*B = I_m - \frac{\mathbf{1}\mathbf{1}^*}{m+1} \preceq I_m.$$

The $m-1$ nonconstant Fourier vectors $(1, \omega^j, \dots, \omega^{(m-1)j})^T$, $1 \leq j < m$, $\omega = e^{2\pi i/m}$, have flat coordinate moduli and span $\mathbf{1}^\perp$. The matrix B preserves their Euclidean norms. Together with $\|z\|_2 \leq m^{1/2-1/p} \|z\|_p$, this proves optimum $m^{1/2-1/p}$ and makes those vectors norming vectors.

For an optimal $Y = B + N$, orthogonality forces N to vanish on every Fourier vector, hence on $\mathbf{1}^\perp$. Therefore $N = u\mathbf{1}^*$ with $u \in \ker A$. Its Gram matrix is

$$Y^*Y = I_m + \left(\|u\|_2^2 - \frac{1}{m+1} \right) \mathbf{1}\mathbf{1}^*.$$

Testing the flat vector $\mathbf{1}$ shows that $\|u\|_2 \leq (m+1)^{-1/2}$ is necessary, and $Y^*Y \preceq I_m$ shows sufficiency. Thus the complete complex minimizer set is

$$\boxed{B + u\mathbf{1}^*, \quad u \in \ker A, \quad \|u\|_2 \leq (m+1)^{-1/2}.$$

This higher-dimensional statement uses complex Fourier vectors. The smallest counterexample is valid over either field.

5 Attribution and verification

The rational verifier checks the right-inverse, Gram, and kernel identities. Continuous parameter ranges and complete minimizer sets are analytic claims. The recovered notes attribute the smallest A, B to Example 4.1 of Dokmanić and Gribonval for the spectral norm; that construction is not claimed as new, and the attribution should be checked before publication.

Recovered references: OpenProblemsInNLA, provisionally IE-23; I. Dokmanić and R. Gribonval, *Beyond Moore–Penrose, Part I: Generalized Inverses that Minimize Matrix Norms*, arXiv:1706.08349v2 (2017), Example 4.1, Corollary 4.2(3), Remark 4.1.¹ The live repository statement was not independently retrieved during reconstruction.

¹<https://arxiv.org/abs/1706.08349>. References retained as source leads; priority and exact source scope require checking.